

XRPL Validator — Daily Engineering Report • 2026-08-28

Compiled retrospectively on 2026-09-23 from the git history. Scope: XRPL work only — the git history of [xrpl-validator](#) and the XRPL Rust SDK, the XRP Ledger itself (amendments) and XRPL research.

Summary

- **6 commits** (xrpl-validator 6)

Engineering changes

- **fix(xrpl-node): probe s2 fallback retries transient failures** (24c2d93 , xrpl-validator) Single-attempt fallback silently kept the primary's lgrNotFound whenever s2 dropped the call (fleet contention: 2428 permanently-uncacheable misses on l106281182, phantom 52/57). Retry 3x like the primary, treat slowDown-class JSON errors as transient, settle on success or a deterministic negative. lgrNotFound from the full-history server itself is settled. DX_RPCLOG gains a FALLBACK-> outcome receipt.
- **feat(xrpl-ledger): BookStep requireAuth — unauth owners reaped, unauth pools yield no synthetic** (46f139b , xrpl-validator) rippled runs requireAuth(book.in, owner) per offer in the stream (BookStep.cpp:755): under the issuer's lsfRequireAuth the OWNER — an AMM pool account included — must hold the ISSUER-side auth flag (issuer low ? lsfLowAuth : lsfHighAuth, the 0x40000/0x80000 line bits) on its line for the asset flowing INTO it; a missing line is tecNO_LINE, and there is NO balance grandfather (that mercy is only DirectStep.cpp:430's ...
- **fix(xrpl-node): probe fallback triggers on 'internal' — rippled masks lgrNotFound for non-admin** (df9a32c , xrpl-validator) the reference node answers the same account_info@pre-window ledger 'lgrNotFound' to an admin curl and 'internal' to the validator host; public s2 answers 'internal' where Clio-backed xrplcluster serves the data. The lgrNotFound-only trigger meant the fallback NEVER fired from the gates — the entire 18-fixture 'deterministic' hist residue was hydration that never got a chance (ours tecNO_DST vs net tecNO_PERMISSION with the dst account provably ...
- **fix(xrpl-ledger): NFT transfer-fee royalty rounds half-even, not floor** (84b245c , xrpl-validator) rippled computes the royalty as multiply(amount, transferFeeAsRate(fee)) (NFTTokenAcceptOffer.cpp:420/:540); asAmount(rate) is non-native so the native fast path never applies and the Number pipeline runs: ONE half-even reduction of the FULL product to 16 significant digits (multiply folds the whole remainder into the tie), /100000 as pure exponent arithmetic, then half-even again at the integer-drop boundary. We ...
- **feat(xrpl-node): validations carry ServerVersion + Cookie; backport kVfFullValidation to canonical tree** (3f9c141 , xrpl-validator) sfServerVersion (UInt64 field 11) encoded per BuildInfo:: encodeSoftwareVersion — 0x183B identifier, 3.3.0, release bit — fills the empty Version column on VHS/xrpscan/livenet. sfCookie (UInt64 field 10): one random per process run, rippled's restart disambiguator. Both sit canonically between SigningTime and LedgerHash.
- **feat(xrpl-node): report our own version — HV 1.0.0, not rippled's** (3f5a198 , xrpl-validator) The ServerVersion identifier becomes 0x4856 (ASCII 'HV', Halcyon Validator) with version 1.0.0: this is an

independent Rust engine tracking rippled 3.3.0's protocol surface, and claiming to BE rippled 3.3.0 misrepresented what runs here. A foreign identifier is inert on-network (BuildInfo::isNewerVersion tests isXrpIdVersion first), and VHS appears to decode semver bits regardless of identifier (44 mainnet ...

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Sponsor-Capped Batch for Game Economies** (gaming) Add a `tfSponsoredBatch` flag plus `sfSponsorID`/`sfSponsorCap` SFields to the outer Batch (XLS-56) transaction; when set, aggregate fees and any own
- **Segregated Sub-Accounts for Atomic DvP** (settlement) Add a new owner-reserved ledger object `ltSUBACCOUNT` (keylet::subAccount(parent, subID)) that holds its own MPToken/RippleState balances and authoriz
- **Unified Content-Addressed Token Metadata** (tokens) Add a reserve-free, content-addressed ledger object `ltMETADATA` (keylet::metadata(sha256(content))) created by a new `MetadataSet` transaction carryi

Appendix — commits

Time	Repo	Commit	Subject
13:26	xrpl-validator	24c2d93	fix(xrpl-node): probe s2 fallback retries transient failures
15:32	xrpl-validator	46f139b	feat(xrpl-ledger): BookStep requireAuth — unauth owners reaped, unauth pools yield no synthetic
18:15	xrpl-validator	df fa32c	fix(xrpl-node): probe fallback triggers on 'internal' — rippled masks lgrNotFound for non-admin
18:41	xrpl-validator	84b245c	fix(xrpl-ledger): NFT transfer-fee royalty rounds half-even, not floor
23:05	xrpl-validator	3f9c141	feat(xrpl-node): validations carry ServerVersion + Cookie; backport kVfFullValidation to canonical tree
23:16	xrpl-validator	3f5a198	feat(xrpl-node): report our own version — HV 1.0.0, not rippled's